High Re Incompressible Flow

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Introduction

The purpose of this assignment is to determine how the two-dimensional, high-Reynolds-number incompressible flow results published by Ghia, Ghia, and Shin ("High-Re Solutions for Incompressible Flow Using the Navier-Stokes Equations and a Multigrid Method," Journal of Computational Physics, 1982) can be reproduced using modern tools, 44 years later. The 1982 paper used a coupled strongly implicit multigrid method to solve the vorticity-stream function form of the incompressible Navier-Stokes equations for the classic lid-driven cavity problem, for Reynolds numbers up to 10,000 and meshes as fine as 257 x 257. This assignment recreates that benchmark problem in Ansys Fluent to (a) remodel the geometry and physics of the paper, (b) demonstrate mesh independence, (c) compare the Ansys results against the published benchmark, and (d) evaluate several turbulence models at Re = 10,000.

Geometry and Setup

The domain re-created from the paper is a 1 m x 1 m square cavity. The top wall (lid) moves at a uniform velocity in its own plane, while the remaining three walls are stationary, no-slip boundaries, the classic shear-driven cavity flow configuration used throughout the original paper. The flow was modeled as two-dimensional, laminar, incompressible flow (with the Reynolds number set directly through the fluid properties and lid velocity), matching the vorticity-transport formulation used in the source paper.

Mesh Independence

To confirm that the solution was not sensitive to mesh resolution, the Re = 400 case was solved on two uniform meshes: a coarser 129 x 129 grid and a finer 257 x 257 grid, mirroring the two mesh densities used in the original paper. For each mesh, the continuity and velocity residuals were monitored until they dropped several orders of magnitude and flattened out, confirming convergence.

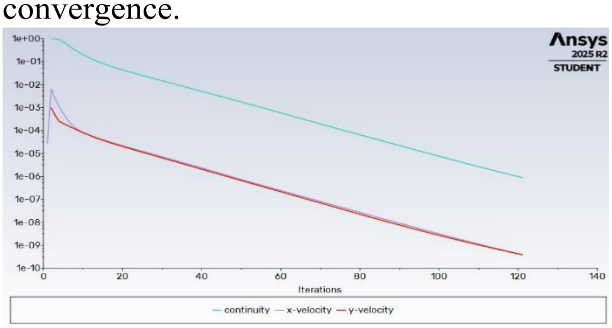

Figure 2a Residual Convergence Re= 400, 129x192 mesh

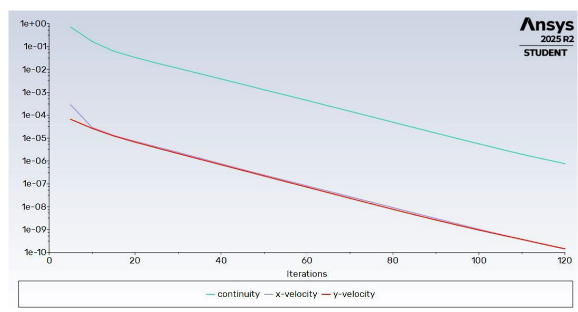

Figure 2a Residual Convergence Re= 400, 257 x 257 mesh

Both meshes converge to residuals on the order of 10^{-6} to 10^{-10} within roughly 100–120 iterations, and the resulting flow fields (shown in Section 4) were visually and quantitatively consistent between the two mesh densities. This confirms mesh independence at Re = 400: the coarser 129 x 129 mesh is adequate for this Reynolds number, consistent with the original paper's own finding that a 129 x 129 grid was sufficient for moderate Re, with the finer 257 x 257 grid reserved for the higher-Re cases where secondary vortices become smaller and more numerous.

Comparison To Paper

Following the mesh independence check, the cavity flow was solved across the same range of Reynolds numbers studied in the original paper: Re = 100, 400, 1000, 3200, 5000, 7500, and 10,000, using the 129 x 129 mesh for the lower Re cases and the 257 x 257 mesh for the higher Re cases, matching the paper's own mesh choices. Table 1 summarizes the cases examined.

Reynolds Number	Mesh 1	Mesh 2	Vortex Pattern
100	129 x 129	—	Single primary vortex
400	129 x 129	257 x 257	Primary vortex + corner eddies
1000	129 x 129	—	Primary vortex + growing corner eddies
3200	129 x 129	—	Additional secondary vortices appear
5000	—	257 x 257	Multiple corner eddies, near-invariant core
7500	—	257 x 257	Multiple corner eddies, tertiary vortices
10,000	—	257 x 257	Fully developed secondary/tertiary vortex system

Table 1. Reynolds numbers and mesh densities used for comparison against the published benchmark.

Re = 100 (129x129 Mesh)

At Re = 100 the flow is dominated by a single, smooth primary vortex centered slightly above and to the right of the cavity center, with no visible secondary vortices in the lower corners. This matches the classic Re = 100 result from the paper, where a single primary vortex was likewise reported with no significant corner eddies at this low Reynolds number.

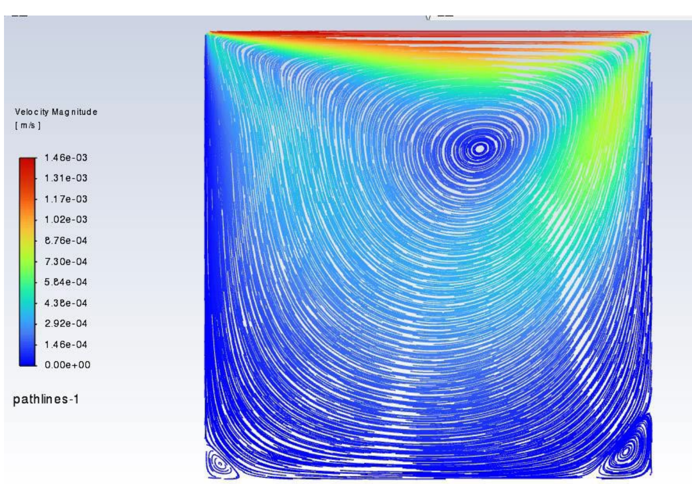

Figure 3: Velocity magnitude path lines Re= 100 (129x129 Mesh)

4.2 Re = 400

At Re = 400, small counter-rotating eddies begin to appear in the bottom-left and bottom-right corners of the cavity, and the primary vortex center shifts toward the geometric center. This behavior, and the corresponding Y-velocity profile shape, were reproduced almost identically between the 129 x 129 and 257 x 257 meshes, reinforcing the mesh-independence conclusion from Section 3, and matches the trend reported in the paper for this Reynolds number.

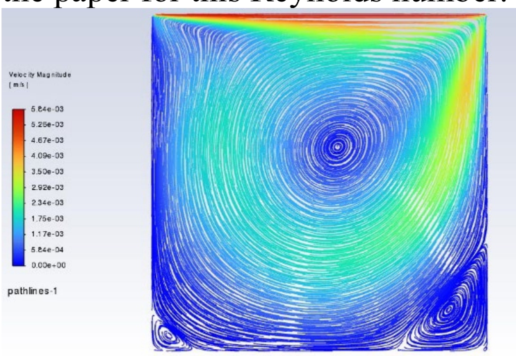

Figure 4a. Pathlines, Re = 400, 129 x 129 mesh.

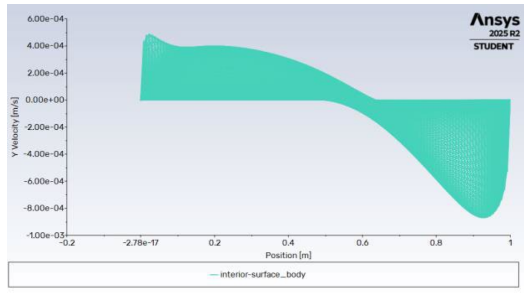

Figure 4b. Y-velocity profile through geometric center, Re = 400, 129 x 129 mesh.

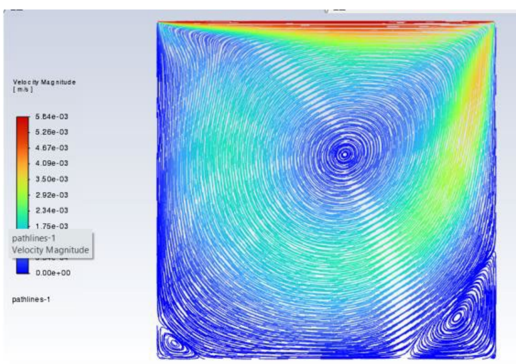

Figure 4c. Pathlines, Re = 400, 257 x 257 mesh.

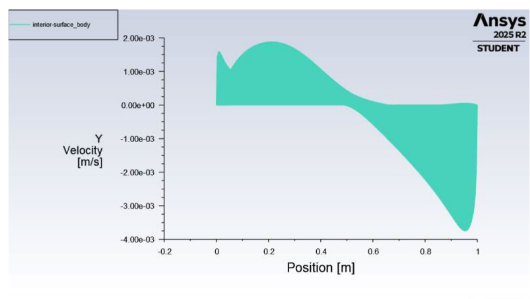

Figure 4d. Y-velocity profile through geometric center, Re = 400, 257 x 257 mesh.

Re = 1000

At Re = 1000, the bottom corner eddies continue to grow and a small counter-rotating vortex becomes visible in the bottom-left corner as well. The primary vortex continues to migrate toward the cavity center, again consistent with the trend documented in the source paper.

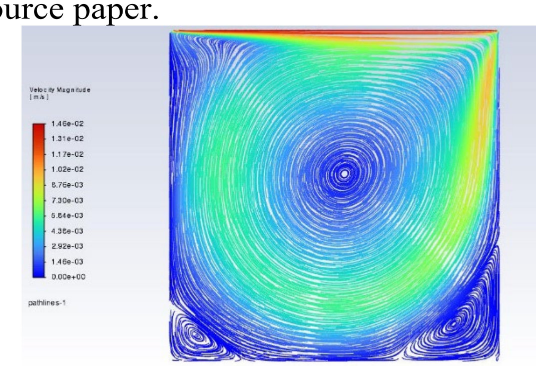

Figure 5a. Pathlines, Re = 1000, 129 x 129 mesh.

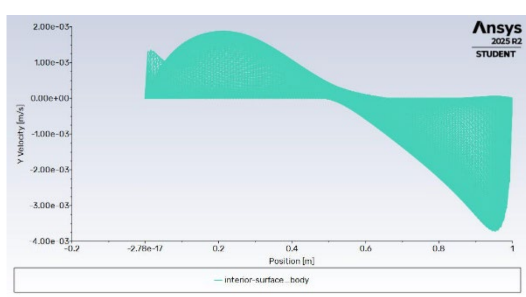

Figure 5b. Y-velocity profile through geometric center, Re = 1000.

Re = 3200

By Re = 3200, additional small vortices appear tucked into the bottom corners beneath the main secondary eddies, indicating the onset of the tertiary corner structures discussed in the paper as Re increases.

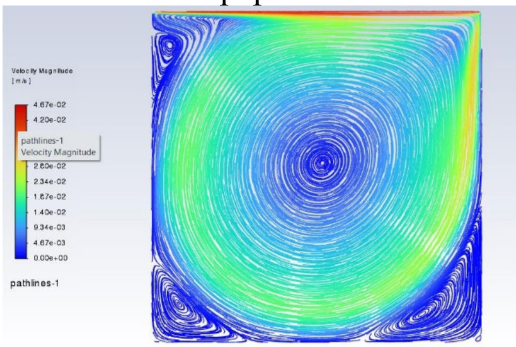

Figure 6a. Pathlines, Re = 3200, 129 x 129 mesh.

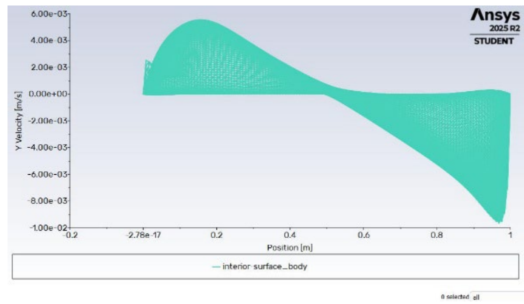

Figure 6b. Y-velocity profile through geometric center, Re = 3200.

Re = 5000 (257 x 257 mesh)

At Re = 5000, the corner vortex system is well developed, with distinct secondary eddies clearly visible in all three non-lid corners. The location of the primary vortex is nearly unchanged from the Re = 3200 case, consistent with the paper's observation that the primary vortex position becomes nearly invariant for Re at or above roughly 5000.

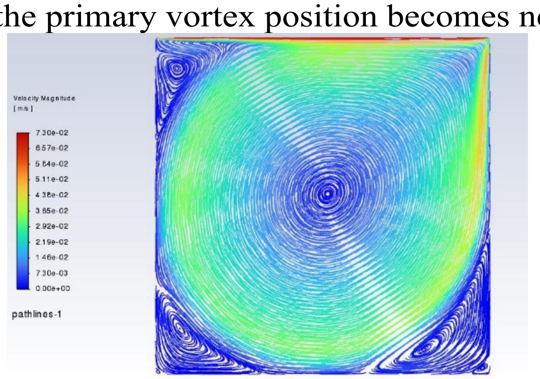

Figure 7a. Pathlines, Re = 5000, 257 x 257 mesh.

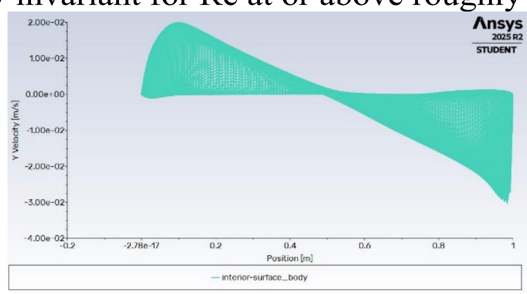

Figure 7b. Y-velocity profile through geometric center, Re = 5000.

Re = 7500 (257 x 257 mesh)

At Re = 7500, the secondary and tertiary corner vortices continue to sharpen and grow slightly, while the primary vortex remains centrally located, again mirroring the trends reported in the paper for this Reynolds number.

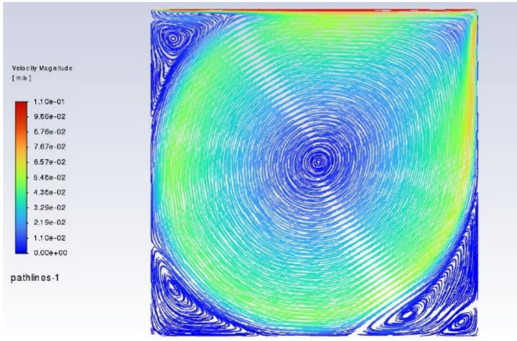

Figure 8a. Pathlines, Re = 7500, 257 x 257 mesh.

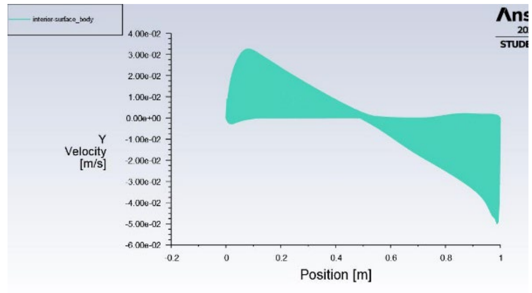

Figure 8b. Y-velocity profile through geometric center, Re = 7500.

Re = 10,000 (257 x 257 mesh)

At Re = 10,000, the highest Reynolds number considered in both the paper and this assignment, the cavity exhibits a fully developed system of secondary and tertiary corner vortices in all three stationary corners, with the primary vortex remaining close to the cavity center. The overall pathline pattern and the shape of the Y-velocity profile closely track the corresponding Re = 10,000 results in the source paper, which the paper's authors obtained using a 257 x 257 grid as well.

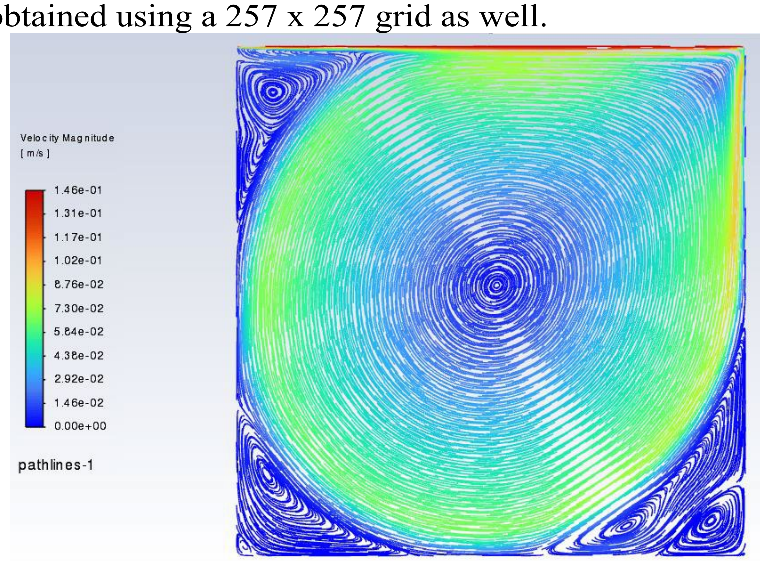

Figure 9a. Velocity-magnitude pathlines, Re = 10,000, 257 x 257 mesh.

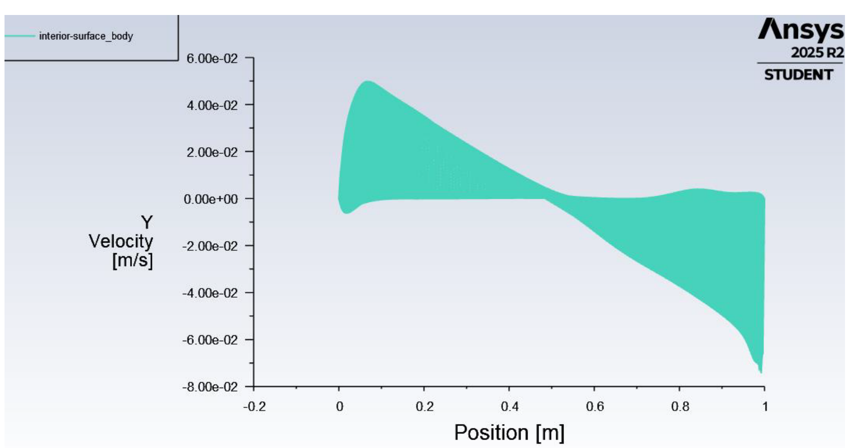

Figure 9b. Y-velocity profile through geometric center, Re = 10,000.

Overall, the Ansys contour and pathline patterns matched the qualitative structure of the published benchmark closely and consistently across the full range of Reynolds numbers, including the appearance, growth, and migration of the primary and secondary corner vortices with increasing Re. This agreement, achieved using a modern general-purpose solver on a laptop-class workstation, is a striking contrast to the original paper's use of a dedicated multigrid scheme on an AMDAHL 470 V/6 mainframe to make the same fine-mesh, high-Re solutions computationally tractable in 1982.

Turbulence Model Comparison at Re = 10,000

For the final part of this assignment, the Re = 10,000 cavity case was re-solved using four different turbulence models available in Fluent: k-epsilon, k-omega, Spalart-Allmaras, and Reynolds Stress Model (RSM). Each model was run to convergence, and the resulting pathline patterns and residual histories were compared for consistency.

k-epsilon Model

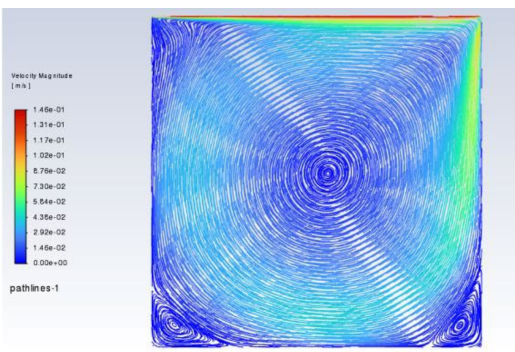

Figure 10a. Pathlines, k-epsilon model.

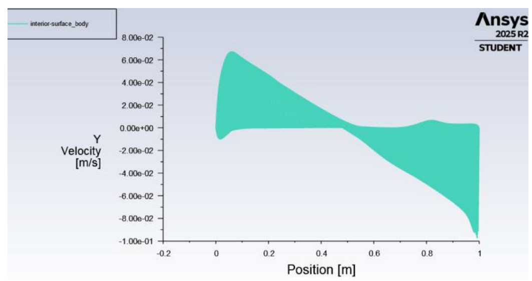

Figure 10b. Residual convergence, k-epsilon model.

k-omega Model

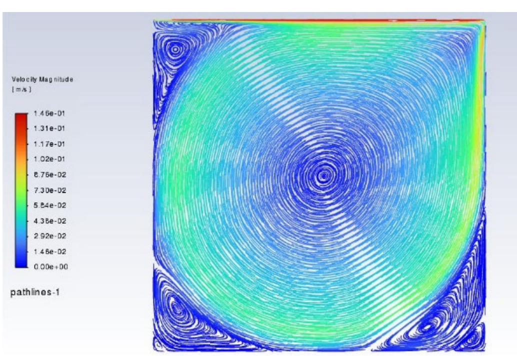

Figure 11a. Pathlines, k-omega model.

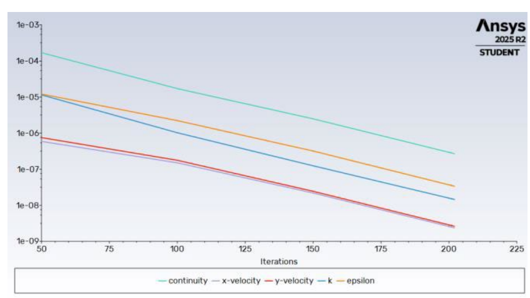

Figure 11b. Residual convergence, k-omega model.

Spalart-Allmaras Model

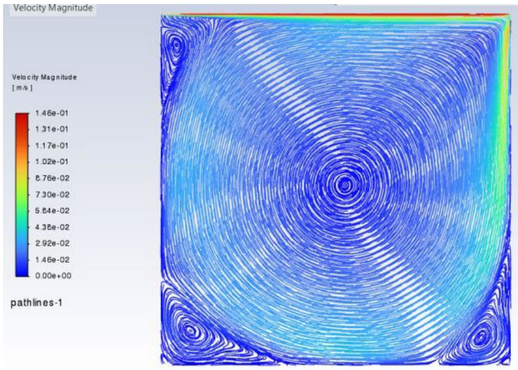

Figure 12a. Pathlines, Spalart-Allmaras model.

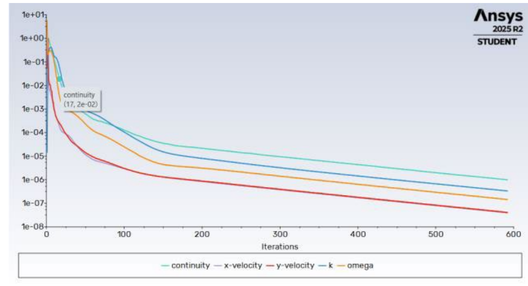

Figure 12b. Residual convergence, Spalart-Allmaras model.

Reynolds Stress Model

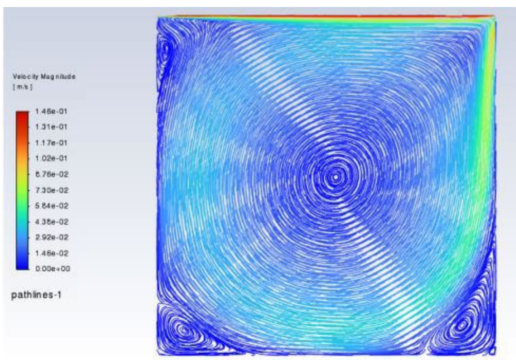

Figure 13a. Pathlines, Reynolds Stress model.

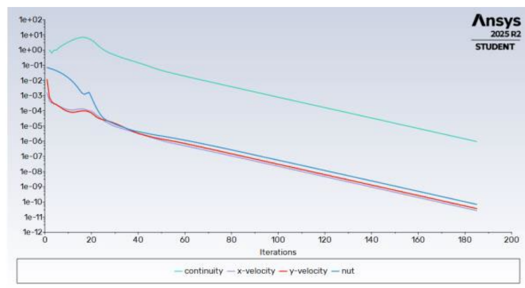

Figure 13b. Residual convergence, Reynolds Stress model.

All four turbulence models produced very similar overall pathline patterns at Re = 10,000, each capturing the primary vortex and the family of corner eddies seen in the laminar high-Re solution. The residual histories for all four models also followed a broadly similar decay trajectory toward convergence, despite the differing number and nature of the transport equations each model solves (e.g., k and epsilon for the k-epsilon model versus the full Reynolds-stress tensor for RSM). This consistency across models suggests that, for this particular flow and Reynolds number, the choice of turbulence closure has a relatively modest effect on the resolved large-scale flow structure, at least at the level of the pathline patterns examined here.

Discussion

This exercise successfully reproduced the qualitative structure of the Ghia, Ghia, and Shin (1982) lid-driven cavity benchmark using a modern CFD package. The mesh-independence study confirmed that a 129 x 129 grid is adequate at moderate Reynolds numbers, while a finer 257 x 257 grid is preferable at higher Re, matching the mesh strategy adopted in the original paper. The comparison across Re = 100 to 10,000 showed the expected progression from a single primary vortex to an increasingly rich system of secondary and tertiary corner vortices, consistent with the published results. It is worth noting, as the original authors also emphasized, that the values of the stream function and vorticity at the vortex centers are considerably more sensitive to mesh resolution than the velocity profiles themselves — a detail worth keeping in mind if a more quantitative, point-by-point comparison against the paper's tables is attempted in future work.

The turbulence model comparison at Re = 10,000 showed reasonable agreement among the four models tested, though this problem is technically laminar-to-transitional at this Reynolds number for a cavity of this size, so the turbulence models are being exercised somewhat outside their typical design envelope; the close agreement between them here should be interpreted as a consistency check on the solver setup rather than a validation of turbulence closure accuracy in a genuinely turbulent flow.

Conclusion

This assignment demonstrated that the fine-mesh, high-Reynolds-number cavity flow results published by Ghia, Ghia, and Shin in 1982 — originally obtained only after significant algorithmic effort (a coupled strongly implicit multigrid method) and non-trivial computation time on a 1980s mainframe — can be reproduced readily with a modern, general-purpose finite-volume solver. The re-created geometry, mesh-independence study, and Reynolds-number sweep all matched the qualitative trends in the original paper, and the four turbulence models tested at Re = 10,000 agreed closely with one another. This highlights just how far commodity CFD tools have advanced in the 44 years since the original study, turning what was once a research-level computational achievement into a routine student assignment.